Exam: F25 Ordinary Differential Equations, Linear Algebra and Vector Calculus

General remarks:

- Read the problems carefully. Only do what you are asked to do.
- You need to show the details of your calculations. It must be clear "What you do" and "How you do it".

Problems:

1) (5% of the total exam)
   Let a be a real number and
   $$A = \begin{pmatrix} 1 & a \\ 1 & 1 \end{pmatrix}.$$
   Calculate all values of a such that the inverse A^{-1} does not exist.

2) (5% of the total exam)
   Let a be a real number and
   $$A = \begin{pmatrix} 1 & a \\ 1 & 1 \end{pmatrix}.$$
   Calculate all values of a such that A has the eigenvalue $\lambda = 1$.

3) (5% of the total exam)
   Consider the linear system of m equations with n unknowns
   $$A\vec{x} = \vec{0}.$$
   Suppose all solutions $\vec{x}$ are of the form
   $$\vec{x} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$
   Find n and $\mathrm{rank}(A)$.

4) (5% of the total exam)
   Consider the nonlinear system of first order ODEs
   $$\begin{aligned} y_1'(t) =& e^{y_1(t)} + y_2(t) \\ y_2'(t) =& e^{y_1(t)} - 2. \end{aligned}$$
   Calculate the location of all critical points.

5) (15% of the total exam)
   Calculate the general solution of
   $$y''(x) - 4y(x) = x + e^x.$$

6) (15% of the total exam)
Solve the initial value problem for $x > 0$

$$2xy(x)^2 + 2x^2y(x)y'(x) = 0, \ y(1) = 1$$

using the theory of exact ODEs.

7) (15% of the total exam)
Solve the initial value problem for $x > 0$

$$xy'(x) + y(x) = 0, \ y(1) = 1$$

using the theory of separable ODEs.

8) (10% of the total exam)
Consider the curve C given by all points $P(x, y, z)$ such that

$$x - y^2 = 0, \ z = x, \ 0 \leq x \leq 1, \ y \leq 0.$$

Find a parametric representation.

9) (10% of the total exam)
Consider the surface S given by all points $P(x, y, z)$ such that

$$x - y^2 = 0, \ z \leq x, \ 0 \leq x \leq 1, \ y \geq 0.$$

Find a parametric representation.

10) (15% of the total exam)
Consider the vector field $\vec{F}(\vec{r}) = (ye^{xy} + ze^x, xe^{xy} + 1, e^x)$. Calculate a scalar field $f(\vec{r})$ such that $\nabla f(\vec{r}) = \vec{F}(\vec{r})$.